

Practice Final F24

Warning: While I attempt to make these close to what the final will be like, it is possible that some topics covered here will not be covered on the final, and some topics not covered here will be on the final.

1. True or False: Prove or disprove the following statements.

(a) $\{T : V \rightarrow V \mid T \text{ is diagonalizable}\} \subseteq \mathcal{L}(V, V)$ is a subspace of $\mathcal{L}(V, V)$

(b) $\{A \in M_{2 \times 2}(\mathbb{R}) \mid \det(A) \geq 0\}$ with the operations of $M_{2 \times 2}(\mathbb{R})$ is a vector space.

2. Suppose we have linear transformations $T_1 : \{0\} \rightarrow U$, $T_2 : U \rightarrow V$, $T_3 : V \rightarrow W$, and $T_4 : W \rightarrow \{0\}$ (U, V, W finite dimensional, $\{0\}$ is the 0-dimensional vector space) where $\text{Ker}(T_{i+1}) = \text{Im}(T_i)$ for all $i = 1, 2, 3$.

(a) Show T_2 is injective and T_3 is surjective.

(b) Show $\dim(V) = \dim(U) + \dim(W)$

(short exact sequence)

3. (a) Suppose β and γ are bases for $P_2(R)$ and R^2 respectively, and $S : P_2(R) \rightarrow R^2$ has matrix representation relative to β and γ given by

$$[S]_{\beta}^{\gamma} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

. Determine if S is injective, surjective, neither injective nor surjective, or an isomorphism.

(b) Suppose now $\beta = \{1, 1 - x, 1 + 2x + x^2\}$ and $\gamma = \left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}$

Compute $[T(x)]_{\gamma}$, then compute $T(x)$

(c) Compute the change of coordinate matrix from γ to the basis $\gamma' = \left\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$ and compute $[S]_{\beta}^{\gamma'}$

4. Let $A = \begin{pmatrix} 1 & -5 \\ 1 & -1 \end{pmatrix} \in M_{2 \times 2}(\mathbb{C})$.

(a) Show that A is diagonalizable and find invertible Q and diagonal D such that $A = QDQ^{-1}$

(b) Compute A^n .

5. 5.2 question 13.

6. (a) Show that for any $b > a \geq 0$, that

$$\langle f, g \rangle := \int_a^b tf(t)g(t)dt$$

defines an inner product on $\mathbb{R}[t]$. HINT: $0 \leq a < t < b$ implies $tf(t)^2 \geq 0$ for $a < t < b$

- (b) If $\int_0^1 tf(t)^2 dt = 4$ and $\int_0^1 tg(t)^2 dt = 9$, then find the range of possible values that $\int_0^1 tf(t)g(t)dt$ can take. HINT: Use the inner product in part a
7. Let $V = P_2(R)$ with inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$
- (a) Starting from the basis $\{1, x, x^2\}$, find an orthogonal basis of $P_2(R)$ with this inner product by applying Gram Schmidt to β .
- (b) Let $W = \text{span}\{1, x\}$. Find orthogonal bases for W and $W^\perp$
8. Suppose that V is an inner product space and $T^*T = I$.
- (a) Suppose that V is an inner product space, $T : V \rightarrow V$ linear, and $T^*T = I$. Prove that T is normal.
- (b) 6.3 Question 13a

Other problems/things of note:

1. 6.1 24
2. 6.2 Q7
3. Relation between having basis of eigenvectors and self adjointness or normalness
4. Suppose T is self adjoint, and λ and μ are two distinct eigenvalues. Prove that $E_\lambda \subseteq E_\mu^\perp$.
5. Self adjoint operators!
Honestly, the first 12 problems in 6.4 except for 7 and 8 look like good practice at first glance if you want additional 6.4 practice.